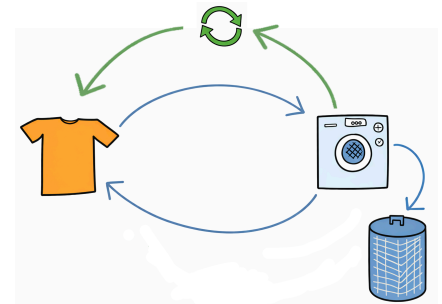


Project description: Textile lifetime analysis

1. What is the overall purpose of the project?

A Danish commercial workwear company operates laundry facilities that process thousands of textile items daily. Each piece of workwear contains a chip with a unique identifier that tracks its journey through the system, recording the product type, how many days it has been in circulation, how many wash cycles it has undergone, and eventually, why it was discarded.

The company wants to understand the lifetime of its textiles. They would like to compare how long clothing lasts across different laundry facilities and between different suppliers. For example, does a t-shirt from Supplier A last longer than one from Supplier B?



The main purpose of our project is to:

- **Assess data quality:** What patterns exist in the data? Are there anomalies, missing values, or inconsistencies that suggest problems with the collection process?
- **Explore what's possible:** What questions can this data reliably answer? What questions remain out of reach given current data limitations?
- **Analyze the lifetime:** By identifying patterns in wear and tear and discards, we can understand why and when textiles are discarded. And, we can help optimize resources and promote a more sustainable textile industry by extending product life.
- **Visualize and communicate:** Create clear visualizations that reveal both expected and unexpected patterns.
- **Recommend improvements:** Based on our findings, how can the company improve its data collection to enable more robust analyses in the future?

2. The Scientific problem or research question

How can we precisely model the true lifespan of industrial textiles when the data contain false administrative registrations, and some of the chips are reused for new customers?

What influence does administrative reasons for discard (e.g. "discontinued model") against physical reasons (e.g. wear and tear) have on the statistical lifespan model?

Is there any significant difference in lifespan between textile type / product type and number of washes?

3. What data are you examining, and what should one pay particular attention to?

We have received 12 datasets from a single laundry facility, one for each month of 2023, detailing the discarded items.

Each observation in the data represents a uniquely identified item and includes both descriptive and lifespan variables. Specifically, the data records the items unique ID, product name, reason for discard, time in circulation, and total wash count.

Pay attention to:

- **Bulk discards:** In 553 rows, the “Pieces of clothing per discard date” is greater than 1. We removed these observations to maintain our focus on individual items, as they constitute a small fraction of the total data.
- **Identity Conflicts:** The unique identifiers are not unique for all articles of clothing, specifically in 439 cases. We removed these cases as they also constitute a small fraction of the dataset.
- **Logical Anomalies:** There are cases where days in circulation are 0, but the total amount of washes is greater than 0.
- **Systematic deviations:** There is a systematic drop off in clothes discarded between year 4-5 and a systematic increase in clothes discarded in year 3 and 5-6.
- **Product Categorization:** The dataset uses different abbreviations for the same type of clothing, so it's important to identify the different abbreviations and group them correctly into standardized clothing categories (e.g., jackets).
- **Weird discards:** There are instances of discards happening after multiple years for reasons that should only realistically be applicable within the first year.

Unik Kode (ui)	Produkt - Produkt
00D007440139	09182 Overall Kblå Hygiejne
00D007440139	23361 Overall Food Kblå

Figure 1: Example of to different types of clothing having the same unique ID

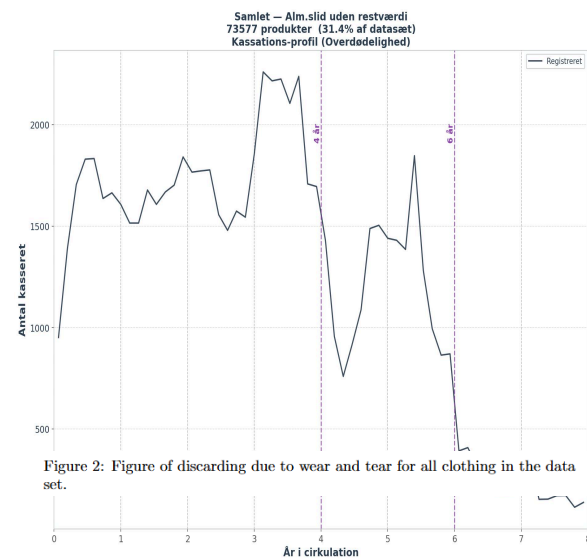


Figure 2: Figure of discarding due to wear and tear for all clothing in the data set.

4. What do you hope to find out?

The goal is to optimize the company's data collection pipeline and utilize this refined data to improve textile sustainability and lifespan. First, we will identify and resolve systemic data anomalies, such as duplicate unique identifiers and illogical inputs, to ensure a robust foundation for future analyses. Then we'll analyse the cleaned dataset to uncover specific durability issues, which will reveal ways for the company to make improvements. To be more specific, we hope to:

- Identifying products that are discarded after very few washes.
- Detect items that systematically underperform compared to similar clothing types.
- Evaluate if specific client industries cause accelerated wear and tear.

The goal is to translate these insights into actionable recommendations that the company can implement. These recommendations should enable the company to take concrete actions, like switching clothing suppliers if the lifetime drops, and being able to continuously compare expected lifetime with the price of textiles.

5. Are there any specific challenges?

A typical facility at the company discards between 10,000 and 30,000 items per month, generating a substantial dataset. However, despite years of data collection, no systematic analysis has ever been performed.

Our main challenge is due to the database not originally being designed for in-depth analysis, leading to inconsistent file naming, questionable structural choices, and vague product descriptions

To address data points where a piece of clothing was discarded with 0 days of circulation, we examined the prevalence of such cases in our data and found 13285 observations. This is quite a substantial amount of observations, but in order to model the lifetime of textiles, we have to remove those observations, as they would introduce downward bias in the estimated average lifetimes.

When tackling the problem that each discarded piece of clothing did not correspond to one row in our data, we found 553 rows where this was the case. This constitutes only 0.22% of our entire dataset and with no direct line of communication with the company with whom the problem could get resolved, we decided to remove those data points in our analysis.

To ensure that we are able to track clothing through time to correctly model the lifetime of the textiles, we need a primary key in the dataset. But we have 439 cases of the unique ID being assigned to multiple pieces of clothing. This could introduce bias in our analysis as different clothes can have very different expected lifetimes, so we decided to remove the duplicates from our dataset as there only were a few.

Regarding categorization of the clothing we have to use string matching as it is not built in as a part of the database. Therefore, we first identified key words and characters in the product names corresponding to a category of clothing. For example T-shirts are written as both “T-shirt” and “Tshirt” and jackets are written as both “jakke” and “jk”. We then handled more specific names of the clothing, i.e. a softshell is also a jacket. We set a threshold value of 2000 to determine whether a clothing will have its own category or would be categorized under another category, i.e. we categorized softshell under jacket. These categories were also saved as separate datasets to limit running time when filtering for a specific type of clothing. This still allows us to explore the lifetime of softshells but also the lifetime of types of clothing with similar use cases efficiently.

A recurring issue in our analysis is whenever we want to dive deeper and try to explain observations we have made with our graph tool, we hit a road block caused by the quality of data and lack of direct contact with the company. For example we tried to look into the systematic discards for t-shirts when they reach 3 and 6 years old. A trend also exhibited by the other categories of clothing but lets stick to the t-shirt example. A large quantity of those discards can be attributed to the single most prevalent product, in this case a white Arla t-shirt, with the biggest reason for discards being customer negligence. 47% of the discards of this product with the before mentioned reason happen at the two time periods and also constitutes a significant portion of the total amount of discards in the spikes. But that is where we hit a wall. We have no contact with the company or Arla, the customer, so we are unable to ask questions that could explain the systematically high amount of discards, and can only speculate as to the reason.

An example of lackluster data is our attempt to model the clothes lifetime as we only have data on discarded clothing and not on the clothing in circulation. As a result we had to model with a Weibull distribution using a Bayesian approach, which is only considered the least worst of the bad models to choose from when we only have the “dead” clothing. We also have only two numerical variables available making the modelling even more challenging. The resulting model was of questionable quality and quite useless from a statistical standpoint as it was unable to capture the systematic variations introduced by the before mentioned spikes meaning that the amount of washes and days in circulation cannot explain the lifetime of clothing fully and other variables are needed to model correctly. This conclusion was based on visual analysis using our graph tool where we regularly saw the model under or over estimate the amount of discards as the spikes caused the mean to be skewed. We also used a chi-squared test to test the model's validity, but as the chi-squared test is quite sensitive to large

quantities of data we got extreme values. But on a positive note, whenever we calculated it on the data subsetting on clothing category and reason for discard the chi-squared values dropped, although never to low enough values to pass the test.

So under these circumstances we were unable to successfully model clothing lifetime as it does not follow the weibull distribution but we had no better alternative distribution to use.

As explained many of our encountered problems have a workaround, but we are still limited in what questions we can answer with the data. The lack of a defined goal before data collection led to a dataset with limited variables of interest. For example, we would have liked to investigate how textile types such as cotton or polyester affect clothing lifetime, but textile type is not a variable in our dataset. Sometimes it is stated within the product name that for example a t-shirt is made of cotton, but that's inconsistent across the dataset, meaning we cannot reliably determine textile type for all observations, thereby making the creation of a lifetime model i.r.t textiles difficult. The effect of textile type on clothing lifetime is one such example of how poor data quality limits the scope of our analysis.

6. What should the final outcome of the project be?

The final goal of the project is to create a model that the company can use to compare the average lifetime of clothes between suppliers, textile and washing facilities. However, as the company is anonymous and the dataset does not include a way of identifying the suppliers nor a way to see what washing facilities were used, we will not be able to test this ourselves and therefore have to set our goal elsewhere.

We therefore hope to get the dataset to a point where we can make assumptions about each clothing type and test these assumptions within reason to get an understanding of what affects the different categories of clothes' average lifetime and why some reasons for discarding are more prominent within some clothing categories. We also hope we will be able to create models that can help compare different styles within each category to each other so it can be used for comparison across suppliers and different clothing textiles.

